GRADIENT DESCENT MANUAL CALCULATION.

Given linear equation: $y = m_1x_1 + m_2x_2 + b$

where : $m = [-1, 2]$

$b = [1, 1]$

$X = [(1,3), (4,10)]$

$y = [5, 6]$

NO1. Computing the predicted values $\hat{y}$ for each data point.

$$\hat{y} = m_1x_1 + M_2x_2 + b \implies \hat{y} = Xm + b \cdot 1$$

$$y = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1\times-1 + 3\times2 \\ 4\times-1 + 10\times2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1+6 \\ -4+20 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

$\therefore \hat{y} = [6, 17]$

Errors: $e = \hat{y} - y$

$$= \begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$\therefore$ According to the errors, data point 1 is off by 1 and data point 2 is off by 11.

Numb 2: Deriving MSE

How much the error changes when we nudge m and b.

Step 1: $\hat{y} = m_1 a_1 + m_2 a_2 + b$

For data point 1: $\hat{y}_1 = 1 \times m_1 + 3 \times m_2 + b$

$\hat{y}_1 = m_1 + 3m_2 + b$

For data point 2: $\hat{y}_2 = 4 \times m_1 + 10 \times m_2 + b$

$\hat{y}_2 = 4m_1 + 10m_2 + b$

Step 2: Change bias into 2x1 column vector.

The bias b is a single number and we have ~~two~~ a column of two number.

- Multiply b by 1.

$b = 1$ Therefore $b \cdot 1 = 1 \times \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Now $b \cdot 1$ is a column with 2 rows, the same shape Xm.

$\therefore \hat{y} = Xm + b \cdot 1$

Step 3: $J = 1/n \sum (\hat{y} - y)^2$ ⟶ Normal formula

But $\hat{y} - y = e$

Hence $e = \hat{y} - y = Xm + b \cdot 1 - y$

Since we have two data points;

$e = [e_1, e_2]$ as a column

$e^T = [e_1, e_2]$ as a row

} This will help with $J = 1/n \sum e^2 = 1/n (e_1^2 + e_2^2)$

$e^T e = e_1 \times e_1 + e_2 \times e_2$

$e^T e = e_1^2 + e_2^2$

$\left(\begin{pmatrix} e_1 \\ e_2 \end{pmatrix} (e_1, e_2) \right)$ e

$J = 1/n \sum (\hat{y} - y)^2$

$= 1/n \sum e^2$

$= 1/n (e_1^2 + e_2^2)$ But $e_1^2 + e_2^2 = e^T e$

$\therefore J = 1/n \, e^T e$.

Step 4: Change in J depends on m_1 and m_2.

$$dJ = \frac{dJ}{dm_1} \times dm + \frac{dJ}{dm_2} \times dm_2 + \frac{dJ}{db} \times db$$

Let $\frac{dJ}{dm_1} = g_1$ and $\frac{dJ}{dm_2} = g_2$

$$\therefore dJ = g_1 dm_1 + g_2 dm_2 + \frac{dJ}{db} db$$

$dJ = [g_1 g_2]\ [dm_1, dm_2] = g^T dm$ — shows how cost changes in relation to m

$$\therefore dJ = g^T dm + \frac{dJ}{db} db$$

step 5: $e = Xm + b \cdot 1 - y$ To see how e changes with change of m, we consider X, y and b to be constants.

$e' = X dm$

But $J = 1/n\ e^T e$

By product rule: $(e^T e)' = (e^T)' e + (e') e^T$

Therefore: $dJ = 1/n\ ((e^T)' e + (e') e^T)$

$= 1/n\ ((e')^T e + (e') e^T)$

But $(e')^T e = e^T e'$ → Because a scalar equals its own transpose

Therefore $dJ = 1/n\ (e^T e' + e' e^T)$

$= 1/n\ (2 (e^T e'))$

$dJ = 2/n\ (e^T e')$

But $e' = X dM$

Therefore $dJ = 2/n\ (e^T \cdot X dM)$

$\frac{dJ}{dM} = \frac{2}{n} \times e^T \times X \frac{dM}{dM}$ → Divide dM both sides

$\frac{dJ}{dM} = \frac{2}{n} \cdot e^T \cdot X$ but $g = \frac{dJ}{dM}$

Therefore $g = \frac{2}{n} \cdot e^T \cdot X$ but $(e^T X)^T = X^T e$ and $e = Xm + b \cdot 1 - y$

$\therefore \frac{dJ}{dm} = g = \left(\frac{2}{n}\right) X^T \cdot (Xm + b \cdot 1 - y)$

step 6: How does e change with when b changes?

$e = Xm + b \cdot 1 - y$ → all except b become constants.

$\therefore e' = 1 \cdot db \Rightarrow de = 1\, db$

But $dJ = \frac{2}{n} e^T e'$ but $e' = 1 \cdot db$

$dJ = \frac{2}{n} e^T \times 1 \cdot db$

but $e^T \times 1 = [e_1\ e_2][1\ 1] = e_1 + e_2$

$dJ = \frac{2}{n}(e_1 + e_2) \cdot db$ → divide by db both sides

$\frac{dJ}{db} = \frac{2}{n}(e_1 + e_2)$ But $e_1 + e_2 = e^T \cdot 1 = 1 \cdot e^T = 1^T \cdot e$

and $e = Xm + b \cdot 1 - y$

$\therefore \frac{dJ}{db} = \frac{2}{n} \cdot 1^T \cdot e$

$= \frac{2}{n} \cdot 1^T \cdot Xm + b \cdot 1 - y$

$\therefore \frac{dJ}{db} = \left(\frac{2}{n}\right) 1^T (Xm + b \cdot 1 - y)$

Step 7: $e = \hat{y} - y = m_1 x_1 + m_2 x_2 + b - y$

$e_1 = m_1(1) + m_2(3) + b - 5$

$= m_1 + 3m_2 + b - 5$

$e_2 = m_1(4) + m_2(10) + b - 6$

$= 4m_1 + 10m_2 + b - 6$

But $J = \frac{1}{n}(e_1^2 + e_2^2)$ But $n = 2$ because of two data points

Hence $J = \frac{1}{2}(e_1^2 + e_2^2)$

By chain rule:

$u = e_1^2 \qquad v = e_2^2$

$du = 2e_1 de_1 \qquad dv = 2e_2 de_2$

$dJ = \frac{1}{2}(du + dv)$

$= \frac{1}{2}(2e_1 de_1 + 2e_2 de_2)$

$= \frac{1}{2} \times 2 (e_1 de_1 + e_2 de_2)$

$= (e_1 de_1 + e_2 de_2)$

But $e_1 = m_1 + 3m_2 + b - 5$ $\quad e_2 = 4m_1 + 10m_2 + b - 6$

$e_1' = 1 \cdot dm_1 + 3dm_2 + 1 \cdot db$ $\quad e_2' = 4 \cdot dm_1 + 10 \cdot dm_2 + 1 \cdot db$

NB: $de_1 = e_1'$

But $dJ = e_1 de_1 + e_2 de_2$

$dJ = e_1 (1 \cdot dM_1 + 3dm_2 + 1 \cdot db) + e_2 (4 \cdot dm_1 + 10 dm_2 + 1 \cdot db)$

$dJ = e_1 dm_1 + 3e_1 dm_2 + e_1 db + 4e_2 dm_1 + 10e_2 dm_2 + e_2 db$

$dJ = (e_1 + 4e_2) dm_1 + (3e_1 + 10e_2) dm_2 + (e_1 + e_2) db$

From step 4 : $\frac{dJ}{dm_1} \times dm_1 + \frac{dJ}{dm_2} dm_2 + \frac{dJ}{db} \cdot db = dJ$

Therefore: $e_1 + 4e_2 = \frac{dJ}{dm_1}$

$3e_1 + 10e_2 = \frac{dJ}{dm_2}$

$e_1 + e_2 = \frac{dJ}{db}$

For $X^T e = [1\ 4; 3\ 10] \cdot [e_1, e_2]$

$= \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 1 \times e_1 + 4 \times e_2 \\ 3 \times e_1 + 10 \times e_2 \end{bmatrix} = \begin{bmatrix} e_1 + 4e_2^2 \\ 3e_1 + 10e_2^2 \end{bmatrix}$

$X^T e = \begin{bmatrix} dJ/dm_1 \\ dJ/dm_2 \end{bmatrix}$

$\frac{dJ}{dm} = \frac{2}{n} X^T e$ $\quad$ If $n = 2$ then $\frac{2}{n} = 1$

$\frac{dJ}{dm} = 1 X^T e$

$\frac{dJ}{dm} = X^T e$

step 8: $\hat{y} = Xm + b \cdot 1 = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$

$e = \hat{y} - y = \begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$

From step 3: $J = 1/n\, e^T e$

$J = 1/n\, e_1^2 + e_2^2$

$= \frac{1}{2} \begin{bmatrix} 1 \\ 11 \end{bmatrix} [1 \;\; 11]$

$= 1/2\,(1 \times 1 + 11 \times 11)$

$= 1/2\,(122)$

$J = 61$

$\frac{dJ}{dm} = X^T e\,(2/n)$ but $n = 2$

$= 2/2 \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 1 \times 1 + 4 \times 11 \\ 5 \times 1 + 10 \times 11 \end{bmatrix} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$

$\therefore \frac{dJ}{dm} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$

$\frac{dJ}{db} = 2/n\, 1^T e = 2/2\,(1 \;\; 1) \begin{pmatrix} 1 \\ 11 \end{pmatrix} = 1 \times 1 + 1 \times 11$

$= 1 + 11$

$= 12$

$\therefore \frac{dJ}{db} = 12$

NO 3. Iteration

$m = m - \alpha \frac{dJ}{dm}$ α = learning rate.

$b = b - \alpha \frac{dJ}{db}$ $\alpha = 0.01$ because the gradients which are 45 and 113 are large, so a small step size keeps the updates from overshooting the minimum.

Iteration 0.

Remember $\frac{dJ}{dm} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$ and $\frac{dJ}{db} = 12$

$$\therefore m = m - \alpha \frac{dJ}{dm}$$

$$= \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.01 \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

$$b = b - \alpha \frac{dJ}{db}$$

$$= 1 - 0.01(12)$$

$$b = 0.88$$

NO 7: i) J falls every time, hence showing that the updates reduce the error

ii) The big drop in Iteration 1 happens because the first gradients were large and positive.